VLSI DESIGN TASK 1

Digital Arithmetic Logic Circuit Library Using Verilog HDL

Prepared for	Industry-Oriented Internship Program on VLSI Design and Semiconductor Engineering
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Repository	github.com/vishnuvamsi396-gif/VLSI-Design-Task-1
Report date	2026-11-01

1. Objective

Implement a small reusable combinational RTL library and demonstrate the Task 1 workflow: digital logic design, synthesizable Verilog, self-checking testbenches, Icarus Verilog simulation, waveform generation, and organized version-controlled documentation.

2. Scope

The project includes basic logic gates, 2:1 and 4:1 multiplexers, half and full adders, a 4-bit ripple-carry adder, a parameterized unsigned adder, and a 4-bit arithmetic/logic mini-project unit. The assignment also introduces Linux-based development and the VLSI design flow; this introductory submission does not claim physical design, synthesis signoff, or chip fabrication.

ASIC and FPGA comparison

Criterion	ASIC	FPGA
Behavior	Fixed after fabrication	Reconfigurable after deployment
Up-front cost	Higher design and fabrication cost	Lower initial cost; no custom mask set
Typical use	Production SoCs and custom accelerators	Prototyping, instrumentation, adaptable networking
Main trade-off	Manufacturing commitment and longer iteration	Programmable resources can cost power, area, or speed

3. Design inventory

Design	Interface / purpose
Basic gates	AND, OR, XOR, NOT, NAND, NOR, XNOR; one-bit truth-table functions.
Multiplexers	2:1 one-bit data selection; 4:1 one-bit selection from a 4-bit input bus.
Half adder	Two one-bit inputs; sum and carry outputs.
Full adder	Two one-bit inputs plus carry-in; sum and carry-out.
4-bit ripple adder	Four cascaded full-adder stages with carry-in and carry-out.
Parameterized adder	Unsigned WIDTH-bit addition with explicit carry-out.
Mini-project unit	4-bit AND, OR, XOR, or addition selected by a 2-bit operation code.

4. Logic reference

The following truth table is the reference for the single-bit gate implementation. A labeled symbol diagram for the basic gates is included as `docs/diagrams/basic_logic_gates.svg` in the repository.

A	B	AND	OR	XOR	NAND	NOR	XNOR
0	0	0	0	0	1	1	1
0	1	0	1	1	1	0	0
1	0	0	1	1	1	0	0
1	1	1	1	0	0	0	1

Boolean expressions

Half adder: SUM = A XOR B; CARRY = A AND B.
Full adder: SUM = A XOR B XOR Cin; Cout = AB + ACin + BCin.
2:1 mux: Y = (~S AND D0) OR (S AND D1).

Karnaugh-map simplifications

Function	Minterms	Minimal SOP
AND	m3	A.B
OR	m1, m2, m3	A + B
XOR	m1, m2	(~A.B) + (A.~B)

Mini-project operation map

operation	Result	Carry-out
00	a AND b	0
01	a OR b	0
10	a XOR b	0
11	unsigned a + b	Addition carry

5. VLSI context

A typical digital IC flow proceeds from specification and architecture through RTL, functional verification, synthesis, implementation (floorplanning, placement, clock-tree synthesis, routing), signoff, and fabrication. ASICs are custom-fabricated for a target product; FPGAs provide reconfigurable programmable logic and are commonly used for prototyping and adaptable products. This repository focuses on introductory RTL and functional simulation.

RTL and simulation workflow

Specification -> RTL modules -> self-checking testbenches -> iverilog compile -> vvp simulation -> pass/fail log and VCD -> GTKWave inspection -> screenshot evidence.

6. Verification plan and observed results

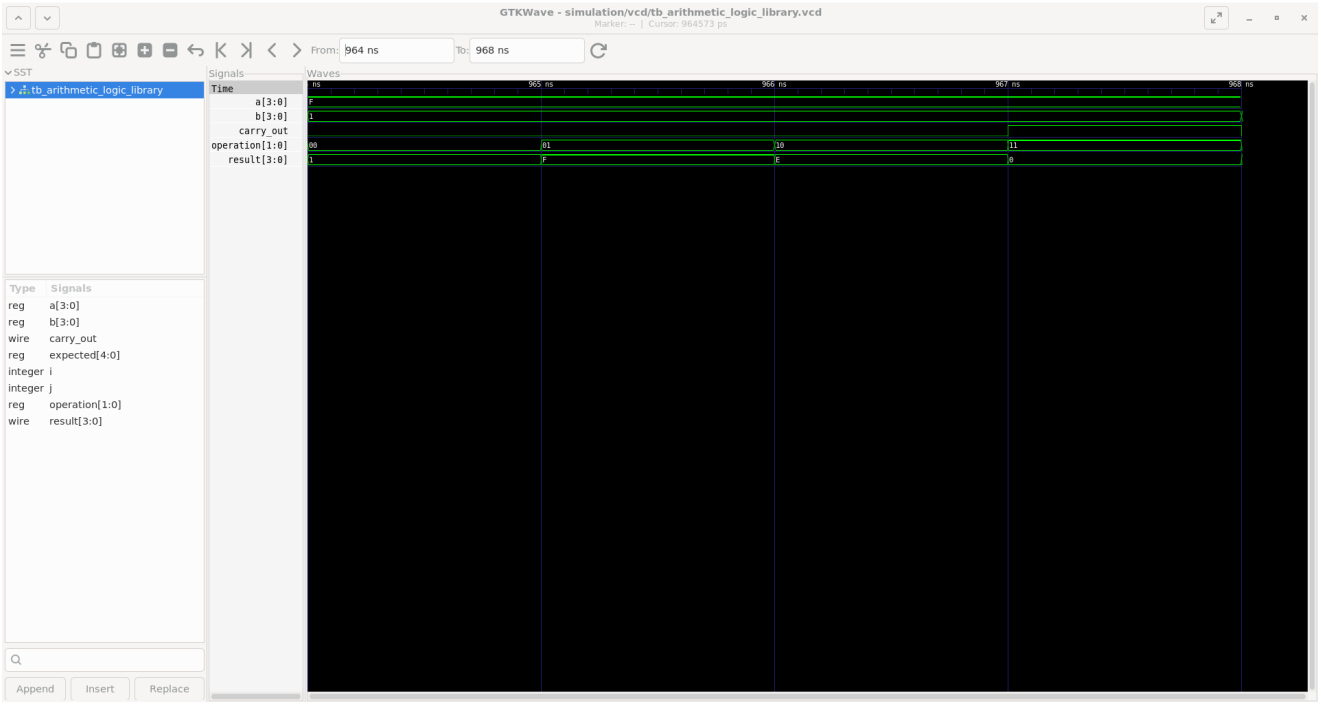
The testbenches use self-checking comparisons and terminate with a fatal error on a mismatch. The suite was run in WSL with Icarus Verilog on 2026-11-01.

Testbench	Coverage	Result
tb_basic_gates	All 4 two-input combinations	PASS
tb_muxes	8 2:1 input/select cases; 4 4:1 selections	PASS
tb_adders	All half-adder and full-adder inputs	PASS
tb_ripple_carry_adder_4bit	All 512 operand/carry-in combinations	PASS
tb_parameterized_adder	1,024 deterministic 8-bit cases	PASS
tb_arithmetic_logic_library	All 4 operations across 16 x 16 operands	PASS

The corresponding simulator logs are in `simulation/results/`; VCD traces are in `simulation/vcd/`. The run script is `scripts/run_all.sh` and returns a failure status if compilation or an assertion fails.

7. GTKWave waveform evidence

The mini-project trace was inspected from 964 ns to 968 ns with a=F and b=1. The operation select steps through AND, OR, XOR, and addition. For operation 11 (addition), the observed result is 0 and carry_out is 1, demonstrating 4-bit unsigned overflow.



8. Environment evidence and conclusion

The student reported Ubuntu 26.04.1 LTS (release 26.04, codename resolute), Icarus Verilog 12.0, VVP 12.0, GTKWave 3.3.126, and Git 2.53.0 on 2026-11-01. The Linux practice report records observed `pwd`, `ls`, and Git commands used to inspect and update the repository. Remaining command exercises are listed as pending until their outputs are recorded.

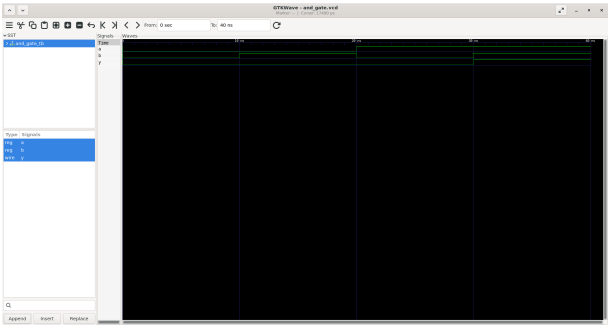
The RTL source and self-checking verification suite are prepared and all six testbenches passed. Seven genuine GTKWave screenshots are included across the mini-project, gates, adders, and 2:1 mux. The repository checklist identifies any remaining student evidence and notes that a LinkedIn post URL, if required by the internship form, must be submitted separately.

Repository checklist

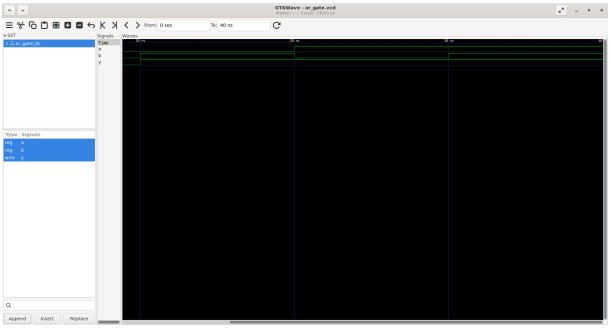
Deliverable	Status
RTL and testbench source	Prepared; all six testbenches passed
Simulation logs and VCD traces	Included from the recorded WSL run
Logic/K-map/design notes and workflow diagrams	Included in docs/
GTKWave screenshot evidence	Seven screenshots: mini-project, gates, adders, and 2:1 mux
Linux environment practice evidence	Ubuntu/tool versions and observed commands recorded; remaining exercises listed

9. Additional GTKWave screenshots

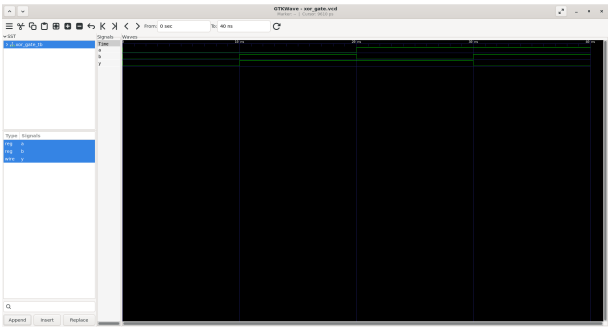
These genuine screenshots were supplied from earlier GTKWave runs and complement the mini-project overflow capture on the previous page.



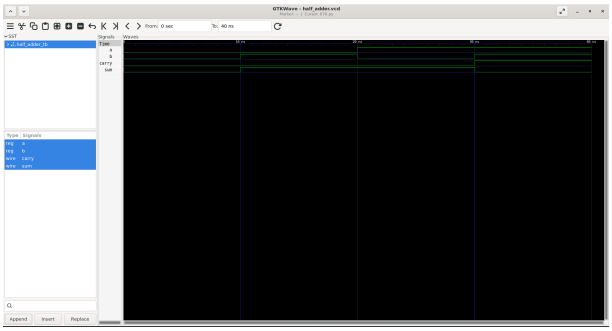
AND gate: a, b, and y across the four input combinations.



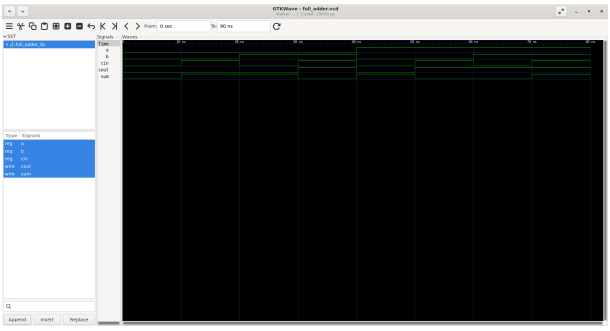
OR gate: output remains high when either input is high.



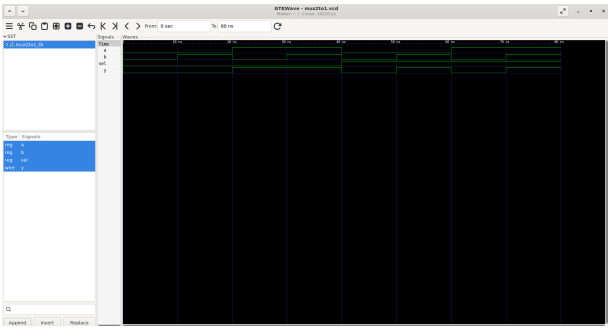
XOR gate: output is high when the inputs differ.



Half adder: sum and carry outputs.



Full adder: sum and carry-out across all input combinations.



2:1 mux: output follows the selected data input.